

Wave 5.2R Stage 9 Temporal Analytical-Residual Models Results

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Summary

Stage 9 completed all 10 first-screen entries without runtime failures. The causal coefficient-residual formulation `k01` is the clear scalar and component leader: it improves raw MAE by 20.53%, mean error by 43.93%, and mean-centered shape error by 9.46% relative to the frozen `h04` analytical anchor.

No candidate passed the complete predeclared gate. The result is therefore a scientifically useful temporal-residual lead, not an official model promotion. `k01` is retained as a qualified research component for later integration and repair work. Stage 10 remains the next roadmap step.

Scope And Controls

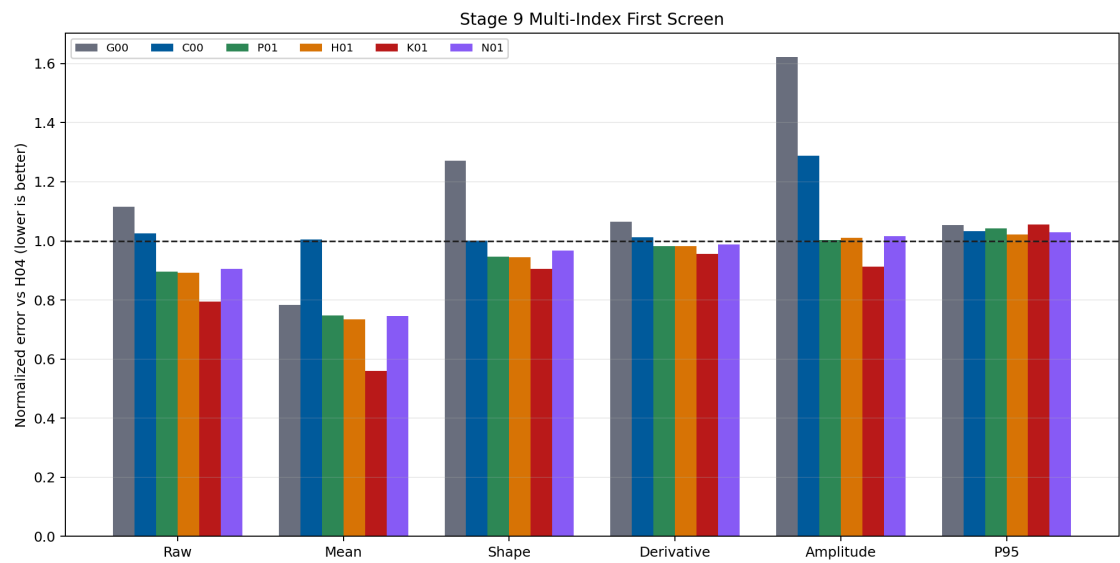
- Dataset: polished dataset, setpoint inputs, forward surface only.
- Split: frozen Stage 0 grouped `675/194/97` train/validation/test split.
- Analytical anchor: frozen Stage 8 `h04`.
- Historical temporal comparator: archived `polished_setpoints_periodic_gru_sequence_fw` checkpoint.
- Temporal contract: unidirectional two-layer GRU with explicit zero-state initialization and state carry across causal chunks.
- First-screen seed: `314159`.
- Target-derived runtime inputs: zero.

The historical `g00` replay was regenerated after detecting that the initial campaign script referenced the older actual-values checkpoint. The corrected replay uses the canonical five-input polished-setpoint forward archive, including the forward direction flag. No candidate was retrained.

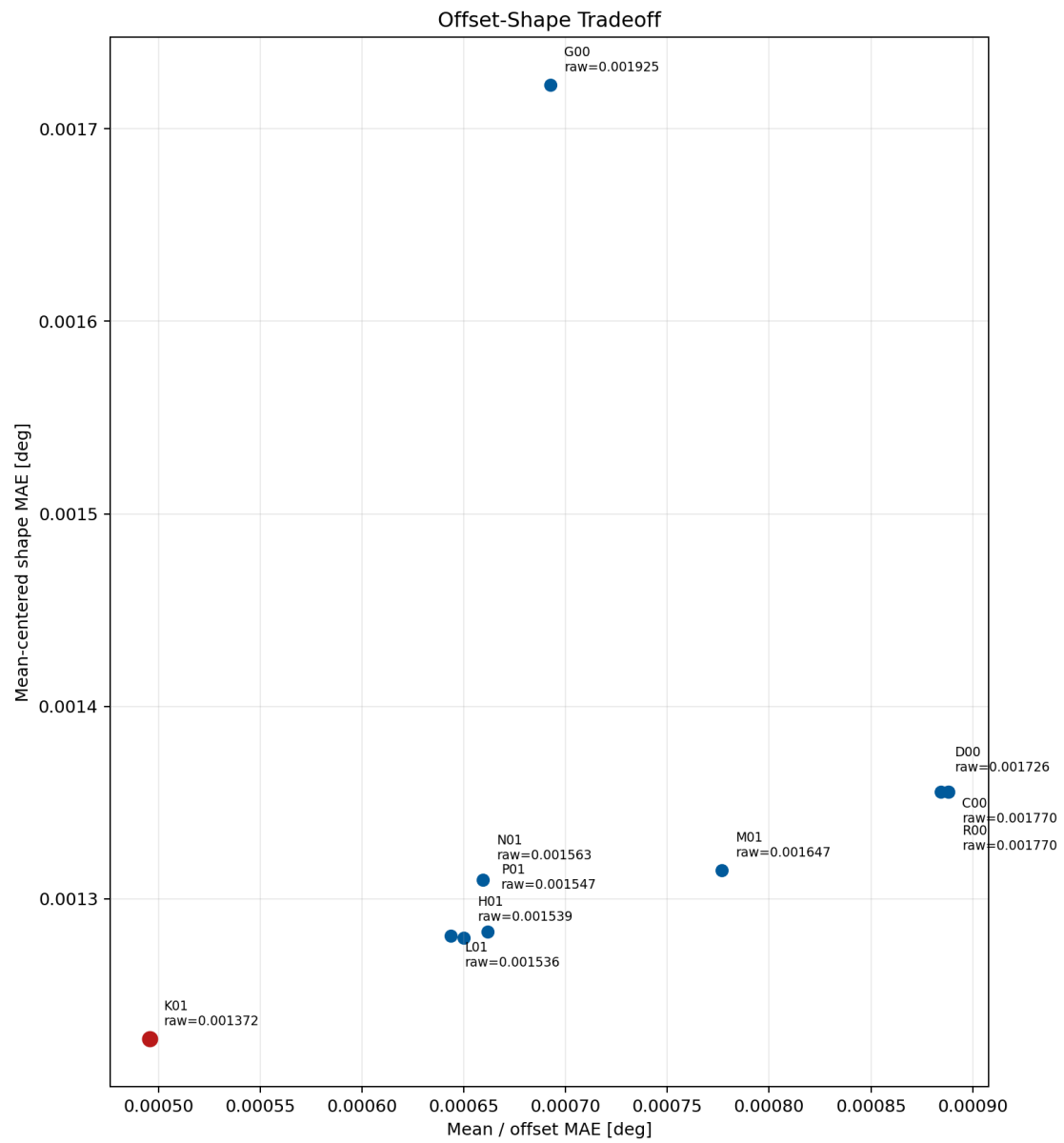
First-Screen Results

Candidate	Formulation	Raw MAE	Mean MAE	Shape MAE	Closure	P95
<code>k01</code>	<code>h04_coefficient_residual_gru</code>	0.001372	0.000496	0.001227	0.000639	0.004148
<code>L01</code>	<code>h04_context_curriculum_residual_gru</code>	0.001536	0.000644	0.001281	0.000463	0.004044
<code>H01</code>	<code>h04_causal_residual_gru</code>	0.001539	0.000650	0.001280	0.000474	0.004017
<code>P01</code>	<code>pf_a_causal_residual_gru</code>	0.001547	0.000662	0.001283	0.000673	0.004097

Candidate	Formulation	Raw MAE	Mean MAE	Shape MAE	Closure	P95
N01	h04_shuffled_angular_order_residual_gru	0.001563	0.000659	0.001310	0.000729	0.004048
M01	h04_static_mean_temporal_shape_gru	0.001647	0.000777	0.001315	0.011127	0.004210
D00	frozen_h04	0.001726	0.000884	0.001356	0.000291	0.003933
C00	causal_periodic_gru	0.001770	0.000888	0.001356	0.013399	0.004066
R00	parameter_matched_zero_anchor_residual_gru	0.001770	0.000888	0.001356	0.013399	0.004066
G00	accepted_setpoint_periodic_gru_replay	0.001925	0.000693	0.001723	0.000234	0.004145



Stage 9 multi-index comparison



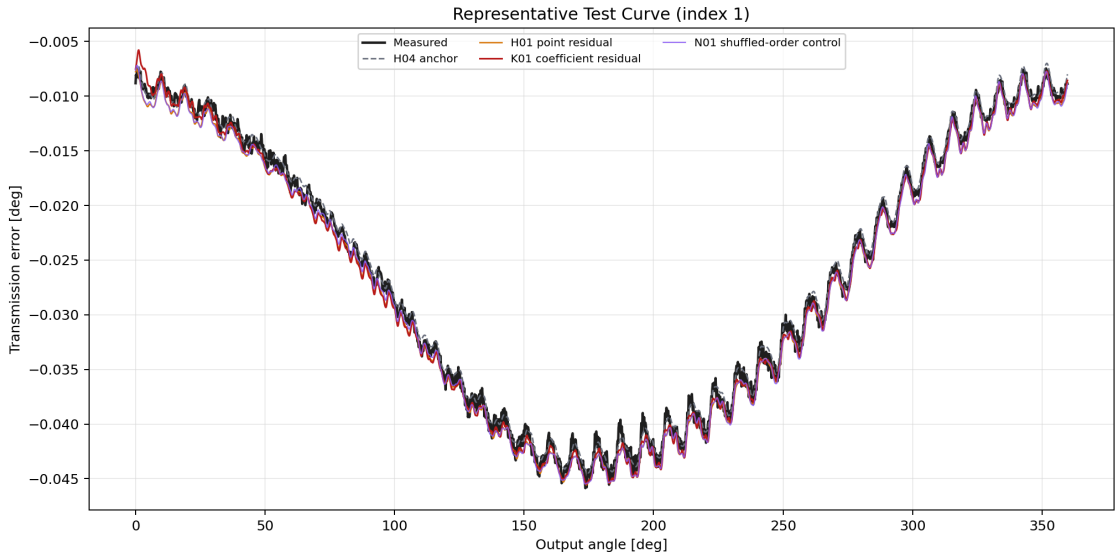
Stage 9 offset-shape tradeoff

What Worked

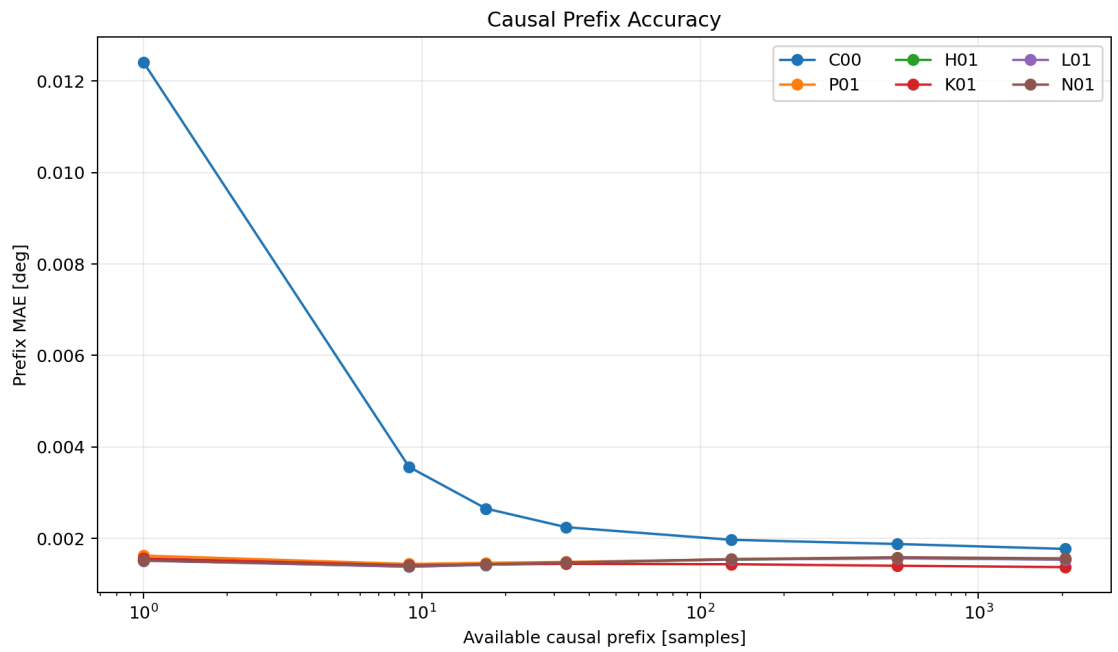
The analytical anchor plus a learned temporal correction is materially better than either constituent alone. **K01** reached raw MAE 0.001372 deg , mean MAE 0.000496 deg , and shape MAE 0.001227 deg . It also beats the corrected accepted GRU replay (0.001925 deg) and the direct causal GRU control.

The coefficient-residual path was stronger than direct point residuals. This supports the premise that the temporal network is most useful when it adjusts an interpretable low-dimensional harmonic representation instead of freely redrawing every angular sample.

H01 , **K01** , and **L01** all beat the shuffled-order control on raw and mean error. Chronological state therefore adds measurable value. However, the shuffled control itself remains strong, showing that a substantial part of the gain comes from angular features, analytical anchoring, and model capacity rather than temporal ordering alone.



Representative measured and predicted curve



Causal prefix accuracy

What Did Not Pass

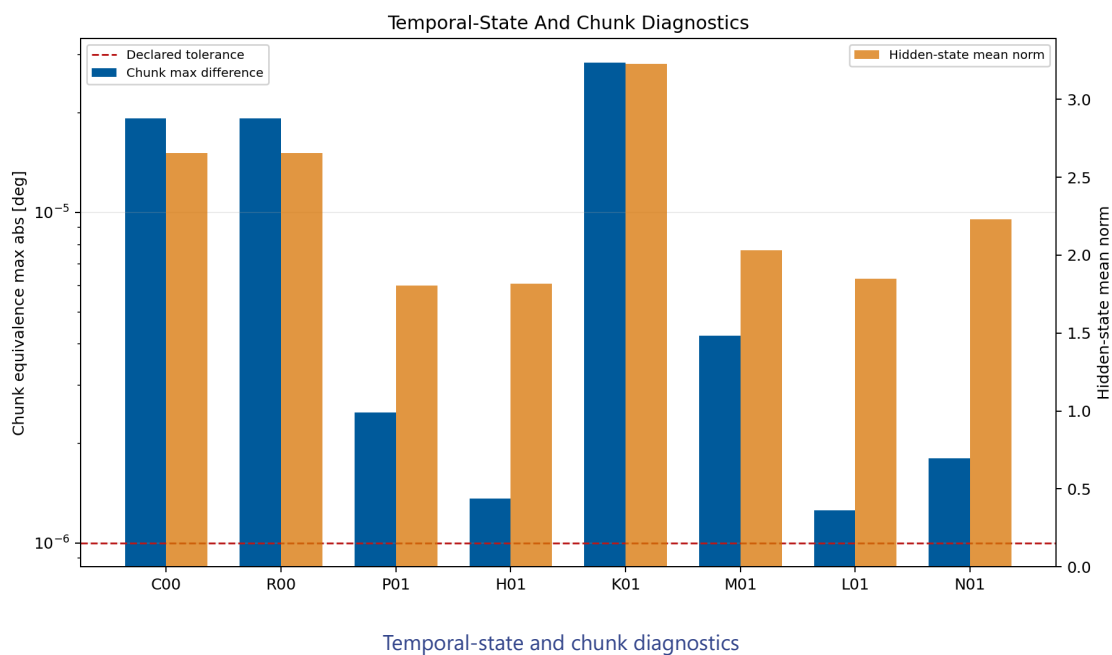
The direct causal GRU control `c00` did not beat `h04` ; its raw MAE was `0.001770 deg` . Temporal memory without the analytical residual structure is therefore not sufficient on this screen.

All physics-guided candidates failed the strict complete gate:

- periodic closure was worse than the best retained baseline;
- per-curve P95 did not remain within the declared 2% tolerance;
- GPU one-pass versus 33-sample chunk evaluation exceeded the predeclared

`1e-6 deg` maximum-difference tolerance. The chunk deviations are small in absolute terms, from approximately `1.25e-6` to `2.84e-5 deg` , and reset reproducibility remained exact. They are consistent with numerical execution-order sensitivity, but the threshold is a predeclared gate and was not relaxed after observing the results.

Temporal-State Diagnostic



Scientific Interpretation

Stage 9 validates the central hybrid-PINN argument: incomplete analytical knowledge can guide the representation while a neural residual compensates for missing effects. The strongest outcome is not a fully free neural sequence model. It is a causal GRU operating on top of the `h04` harmonic anchor and modifying harmonic coefficients.

The result is nevertheless not deployment-ready. The average behavior improves strongly, while tail curves and periodic boundary consistency regress. Future use of `k01` should therefore preserve its coefficient-residual structure but add explicit boundary-consistent parameterization, tail-risk selection, and a numerically calibrated chunk-equivalence audit.

Decision

- Stage 9 status: completed without promotion.
- Official promoted candidate: none.
- Retained research component: K01 .
- Stability repeats: not launched because no candidate passed every first-screen gate.
- Next roadmap action: Stage 10 sparse and symbolic formulation discovery.

Reproducibility Evidence

- Campaign leaderboard:
output/training_campaigns/2026-07-29-18-52-55_wave52r_stage9_temporal_analytical_residual_models_2026_07_29/campaign_leaderboard.yaml
- Gate summary:
output/training_campaigns/2026-07-29-18-52-55_wave52r_stage9_temporal_analytical_residual_models_2026_07_29/campaign_first_screen_gate_summary.yaml
- Exit-gate summary:
output/analysis/wave_5_2r/stage9_temporal_analytical_residual_models/closeout/stage9_exit_gate_summary.yaml
- Accepted GRU replay:
output/analysis/wave_5_2r/stage9_temporal_analytical_residual_models/stage9_accepted_gru_replay.npz